SPATIAL AFFORDANCE TOKENIZATION VS. PASSIVE DIGITAL CLASSIFICATION

Transforming 2D camera pixels into metric SE(3) coordinates, contact normals, and physical affordances

PASSIVE DIGITAL CLASSIFICATION (DISEMBODIED ML)

`2D Pixels` RGB (H×W×3) → `ResNet / CLIP` Global Pooling → `"mug": 0.94` Abstract Text Label

❌ **Zero Metric Scale:** Cannot deduce distance in meters or gripper aperture.
❌ **Zero Kinematic Utility:** Motor controller cannot actuate on a semantic string.
❌ **Blind to Geometry:** Ignores surface normals, collision hulls, and center of mass.

SPATIAL AFFORDANCE TOKENIZATION (PHYSICAL AI)

`RGB-D + PTP` Synchronized → `ViT / DINOv2` Patch Unproject → `SE(3) Tokens` Metric 3D Geometry

✓ **Metric Grounding:** Explicit coordinate frames relative to base frame (*T_base_cam*).
✓ **Direct Kinematic Authority:** Feeds inverse kinematics and trajectory decoders.
✓ **Mathematical Safety Barrier:** Bounding hulls supply distance *d_clear* to MCU CBF.

THE 4 ESSENTIAL PHYSICAL AFFORDANCE TOKENS EMITTED BY EMBODIED PERCEPTION

1. METRIC GRASP POSE (SE(3))

`T_grasp = [ R | p ]`

- Translation *p = [x, y, z]^T* (m)
- Orientation *q = [qw, qx, qy, qz]*
- Gripper width aperture *w_grip*

```
x = +0.342 m ± 0.004
y = −0.118 m ± 0.003
z = +0.085 m ± 0.002
```

2. CONTACT SURFACE GEOMETRY

`n̂ = [nx, ny, nz]^T, κ`

- Unit surface normal vector *n̂*
- Local principal curvature *κ*
- Approach alignment axis *v_app*

```
n̂ = [0.02, 0.98, −0.19]
Approach: ∡(v, n̂) < 5°
Contact patch: Flat
```

3. MATERIAL & FRICTION TENSOR

`μ_est, K_stiff, m_est`

- Friction coefficient *μ̂ ≈ 0.45*
- Material compliance *K_stiff*
- Estimated mass *m_payload*

```
μ = 0.52 (Ceramic)
F_normal_min = 4.2 N
Payload: 0.380 kg
```

4. OBSTACLE BARRIER ENVELOPE

`d_clear(x) ≥ d_safe`

- Dynamic bounding box *B_obs*
- Velocity vector *v_obs*
- Direct feed to MCU CBF

```
d_clear = 0.142 m
d_stop = 0.088 m
Safety Margin: +54 mm
```